FastSpeech 2 Text-to-Speech for English and Telugu

Technical report · two separately trained models (Svarah English, IndicTTS Telugu) · November 2026

Summary. Two FastSpeech 2 models were trained from scratch, one per language, starting from the ming024/FastSpeech2 PyTorch implementation. Each uses about 5 h of training speech and 1 h of validation speech, with no sentence shared between the two splits. Both models train stably over 60,000 steps (2.0 h for English, 2.5 h for Telugu on one RTX 5060 Ti). The Telugu model produces natural, fluent speech: on 100 held-out sentences an ASR model (Whisper large-v3-turbo) recognises the synthesised audio as well as the real recordings (character error rate 15.8 % vs 16.2 %). The English model is clearly weaker (word error rate 18.2 % vs 5.7 % on real recordings) because Svarah offers only about 2 minutes of mostly spontaneous, non-studio speech per speaker. Choosing the most intelligible of the 115 learnt English voices roughly halves the error on the test sentences.

	English (Svarah)	Telugu (IndicTTS Telugu)
Training / validation speech	5.05 h (3,590 utt.) / 1.05 h (790 utt.)	5.04 h (2,398 utt.) / 1.02 h (485 utt.)
Speakers	115 (median 1.9 min of training speech each)	2 (one female, one male, 2.5 h each)
Text front-end	number/time normaliser, LibriSpeech lexicon + g2p_en, POS-based heteronyms	rule-based G2P over Telugu script, Telugu number words
Aligner	MFA 3.4.3, pretrained `english_us_arpa`	MFA 3.4.3, acoustic model trained on the selected data
Best validation loss (step)	2.614 (13k); 2.650 at 60k	1.960 (32k); 1.971 at 60k
Submitted checkpoint	step 60,000	step 32,000 (best validation)
ASR error on 100 validation sentences: synthesised / real	WER 18.2 % / 5.7 % (CER 9.8 % / 2.8 %)	CER 15.8 % / 16.2 % (WER 69.4 % / 67.0 %)

1. Implementation approach

1.1 Model components taken from the reference implementation

The model is the FastSpeech 2 of Ren et al. (2021) as implemented in `ming024/FastSpeech2`; the architecture and losses were kept unchanged so that the two languages differ only in data, front-end and configuration.

- **Encoder**: phoneme embedding (256-d) + sinusoidal positions, 4 feed-forward Transformer blocks (2-head self-attention, 1D-conv feed-forward with kernels 9 and 1, 1024 filters, dropout 0.2).
- **Speaker embedding** (256-d) added to the encoder output; 115 speakers for English, 2 for Telugu.
- **Variance adaptor**: duration, pitch and energy predictors (each two 1D conv layers with 256 filters, kernel 3, ReLU, layer norm, dropout 0.5, and a linear output). Pitch and energy are predicted per phoneme, quantised into 256 linear bins between the training minimum and maximum, embedded and added to the hidden sequence. The length regulator repeats every phoneme state by its duration in frames. During training the ground-truth duration, pitch and energy are fed to the adaptor; at inference the predictions are used.
- **Decoder**: 6 feed-forward Transformer blocks, a linear projection to 80 mel bins, and a 5-layer convolutional PostNet (an addition of the reference implementation, not in the paper).
- **Losses**: L1 on the mel spectrogram before and after the PostNet, MSE on phoneme-level pitch and energy (normalised), MSE on log(duration + 1). The total loss is their unweighted sum.
- **Optimiser**: Adam ($\beta = 0.9, 0.98$, $\varepsilon = 10^{-9}$) with the Transformer warm-up schedule (peak learning rate $\approx 1\times10^{-3}$ at step 4,000) and gradient clipping at 1.0.

- **Vocoder**: the pretrained universal HiFi-GAN shipped with the reference (22,050 Hz, 80 mel bins, f_{max} 8 kHz). It is not fine-tuned.
- **Features**: STFT with 1024-point window and 256-sample hop (11.6 ms frames); F0 from WORLD (DIO + StoneMask), linearly interpolated through unvoiced frames and averaged per phoneme; energy as the L2 norm of each STFT frame, averaged per phoneme; durations from forced-alignment TextGrids.

The paper predicts frame-level pitch through a continuous wavelet transform; the reference implementation (and this work) instead predicts one F0 value per phoneme, following FastPitch, which the reference's notes report to give better prosody. This is why the pitch and energy plots in this report are step functions over phonemes.

1.2 Code organisation

The required layout is kept (`English/` and `Telugu/` each with `preprocess.py`, `train.py`, `inference.py`, `config.yaml`, `checkpoints/`). The model code is shared in `fastspeech2/` rather than duplicated. Each language has a single `config.yaml` holding the reference's `preprocess.yaml`, `model.yaml` and `train.yaml` as three sections. Small language files (`metadata/`: speaker table, pitch/energy statistics and the exact train/validation file lists) are kept in the repository so that inference runs without the corpora.

1.3 Modifications to the reference implementation

File	Change
Shared (both languages)	
`text/telugu.py` (new)	Telugu front-end: Unicode normalisation, Telugu digits, number/ordinal/time expansion into Telugu words, rule-based grapheme-to-phoneme conversion (section 3.4).
`text/english.py` (new)	English front-end: times, dates, ordinals, years and numbers to words; lexicon lookup with g2p_en fallback; part-of-speech tagging to choose between heteronym pronunciations (*conduct* noun vs verb).
`text/symbols.py`	46 Telugu phones (12 vowels, 34 consonants, as lower-case ASCII labels) added to the symbol table next to ARPAbet; the pinyin set removed.
`preprocessor.py`	Train/validation split read from the selection step instead of a random split after feature extraction; pitch/energy statistics from the training split only; reproducible speaker ids; parallel feature extraction; MFA 3 silences (empty intervals) read correctly; utterances containing `spn` (unalignable speech) dropped; all pauses mapped to one `sp` symbol; a frame limit of 1,300 (15 s).
`trainer.py`, `evaluate.py`	Single-GPU loop, optional bf16 autocast (≈1.5× faster), fixed seed, CSV loss logs, best-validation checkpoint kept as `best.pth.tar`.
`model/modules.py`	Bug fix: the energy predictor received the pitch control factor; length regulator vectorised with `repeat_interleave`; tensors created on the model's device.
`audio/*`, `utils/*`, `model/optimizer.py`	Compatibility with librosa 0.11, NumPy 2 and PyTorch 2.11 (keyword arguments, float32 energies, `weights_only` checkpoint loading); vocoder path relative to the package; float16 weights-only checkpoint export.
`synthesis.py`, `plot_losses.py`, `asr_eval.py` (new)	Per-phoneme duration/F0/energy predictions with the three plots and a JSON file per sentence; loss plots; ASR-based intelligibility evaluation.
English only	
`English/preprocess.py`	Reads the Svarah parquet files; derives speaker ids from the speaker metadata (the corpus has no speaker column); filters transcripts with tags or ambiguous symbols; per-speaker proportional selection; MFA dictionary from the LibriSpeech lexicon plus g2p_en; alignment with the pretrained US-English MFA model.
`English/inference.py`	Default voice chosen by ASR intelligibility (section 5.2).
Telugu only	

`Telugu/preprocess.py`	Reads the IndicTTS parquet files; speaker-balanced selection grouped by sentence; lexicon generated by the rule-based G2P; a Telugu MFA acoustic model trained on the selected data and then used to align it (no pretrained Telugu aligner exists).

2. Datasets, selection and preprocessing

2.1 The corpora as found

Svarah (AI4Bharat) is an Indian-accented English benchmark built for speech recognition: 6,656 utterances, 9.6 h at 16 kHz, from 117 speakers across many Indian states, a mix of read sentences and spontaneous answers recorded outside a studio. There is no speaker-id column and the file names identify recording sessions (2,938 distinct prefixes), not speakers. The eight speaker-metadata columns (gender, age group, first language, state, district, education, job category, occupation domain) take 115 distinct combinations, which were used as speaker ids (at most two pairs of the 117 speakers are merged).

IndicTTS Telugu (IIT Madras, Hugging Face copy) is studio speech from two professional voice artists. The files are 16 kHz and total 36.7 h in 8,576 clips, although the dataset card states 48 kHz and 8.7 h, so all numbers here were measured from the audio. Most female clips are whole paragraphs (median 18.5 s); 632 male clips repeat sentences that the female speaker also read. A few transcripts type a Latin "o" in place of the visually identical anusvara ◌ం, which the normaliser repairs.

2.2 Selection and subsampling

Quotas are counted in *speech* hours: each clip's duration after trimming leading and trailing silence (librosa, 30 dB below peak). This was calibrated against the final MFA alignments (Telugu within 0.4 %, English 5 % high, hence the selection margins of 1.02 and 1.06). Clips shorter than 1 s or longer than 15 s of speech were excluded; 15 s also bounds the decoder's sequence length.

- **Telugu**: transcripts that still contain Latin letters after normalisation were removed (5,496 usable clips, 11.1 h of speech). Clips were grouped by normalised sentence text and the groups shuffled with a fixed seed (1234). Groups were assigned first to validation, then to training, until each speaker reached half of the split's quota: 0.51 h per speaker for validation and 2.55 h per speaker for training.
- **English**: transcripts with annotation tags or symbols whose spoken form is ambiguous (such as `[ ] < > # * / = + % & $`, phone-number-like digit strings, numeric ranges) were removed (4,823 usable clips, 7.05 h of speech). Each speaker's quota is proportional to how much usable speech the speaker has. This keeps all 115 speakers in training and 114 in validation (every validation speaker is also a training speaker, so the speaker embedding is learnt), and keeps the 5:1 ratio for every speaker. A clip is taken only if it brings the speaker closer to its quota, which avoids a systematic overshoot of the small (≈30 s) validation quotas.

2.3 Train / validation split

The split is decided once, at selection time, and written to `split.json`. Because whole sentence groups go to one split, the two splits share no sentence text: overlap 0 for both languages, checked in `selection_summary.json`. Validation speech is never used for training or for the pitch/energy normalisation statistics; it only guides checkpoint and voice selection. The final lists are `metadata/train.txt` and `metadata/val.txt`.

Stage	English utt.	English hours	Telugu utt.	Telugu hours
Corpus	6,656	9.61	8,576	36.71
Usable after filters (speech hours)	4,823	7.05	5,496	11.10
Selected: train / val	3,600 / 792	5.33 / 1.11*	2,418 / 488	5.10 / 1.02
After alignment and feature extraction: train	3,590	5.05	2,398	5.04
After alignment and feature extraction: val	790	1.05	485	1.02

* Trim-based estimate including the 6 % margin; final hours are frame counts between the first and last aligned phone. Lost clips failed alignment, contained `spn`, or exceeded 1,300 frames.

2.4 Preprocessing

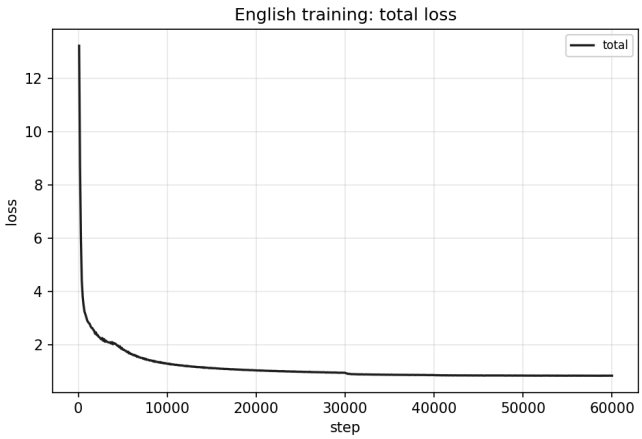
1. **Audio**: decoded from the parquet files, mixed to mono, resampled from 16 kHz to 22,050 Hz for the vocoder, and peak-normalised to 0.95 (the reference scaled to exactly 32,768, which overflows 16-bit PCM). Since the sources are 16 kHz there is no content above 8 kHz, which matches the mel f_{max} of 8 kHz.
2. **Telugu text**: NFC normalisation; Telugu digits to ASCII; zero-width joiners removed; Latin "o" after Telugu letters restored to an anusvara; times (10:45 → పది నలభై ఐదు), ordinals (12వ → పన్నెండవ) and numbers (2026 → రెండు వేల ఇరవై ఆరు) spelled out. The G2P walks the aksharas: a consonant takes the following vowel sign, loses its vowel before a virama, and otherwise keeps the inherent /a/; the anusvara becomes the nasal homorganic with the next consonant (ng, nj, nx, n) and /m/ elsewhere; the visarga becomes /h/. Retroflex consonants are written with an x (tx, dx, nx, sx, lx). Telugu spelling is close to phonemic, so the rules cover nearly all words; the generated lexicon has 13,629 entries.
3. **English text**: times (10:45 AM → "ten forty five A M"), day-month dates with ordinals ("twelfth November"), years ("twenty twenty six") and other numbers are expanded. Words are looked up in the LibriSpeech lexicon of the reference, then the CMU dictionary, then predicted by g2p_en (1,083 of 7,112 corpus words). At synthesis time NLTK part-of-speech tags select between heteronym pronunciations.
4. **Alignment**: Montreal Forced Aligner 3.4.3. English uses the pretrained `english_us_arpa` model, whose phone set matches the ARPAbet symbols of the reference. Telugu has no pretrained model, so MFA's default training recipe (monophone up to speaker-adapted triphone models) is run on the selected 6 h, which are then aligned with the resulting model. 4,382 of 4,392 English and 2,883 of 2,906 Telugu clips aligned.
5. **Features**: mel spectrogram, phoneme-level F0 and energy, and durations (frames per phone) as in the reference. F0 and energy are standardised with the mean and standard deviation of the training split.

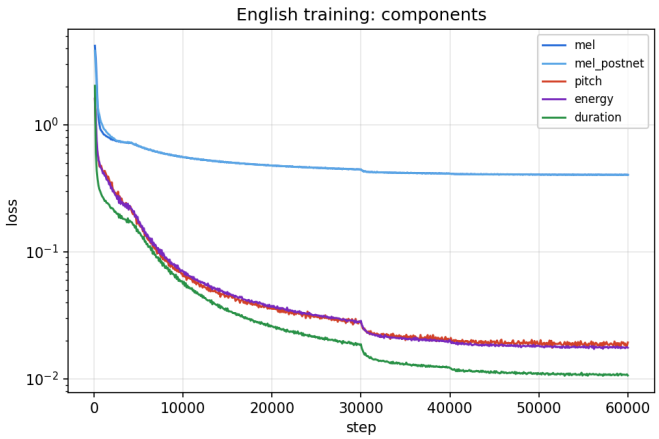
3. Training configuration

Setting	Value (both languages)
Steps, batch size	60,000 steps, batch 16 (sorted into length groups of 4 batches as in the reference)
Learning rate	warm-up 4,000 steps; multiplied by 0.3 at 30k, 40k and 50k (the reference's 300k/400k/500k schedule scaled to the shorter run)
Regularisation	dropout 0.2 in encoder/decoder, 0.5 in variance predictors; gradient clipping 1.0
Precision, seed	bf16 autocast for the forward pass, losses in float32; seed 1234
Validation, checkpoints	full validation set every 1,000 steps; checkpoints every 10,000 steps and on every new best validation loss
Parameters	35.30 M (English, with 115 speaker embeddings), 35.27 M (Telugu)
Hardware, time	NVIDIA RTX 5060 Ti 16 GB; English 2 h 00 min, Telugu 2 h 32 min (longer utterances)

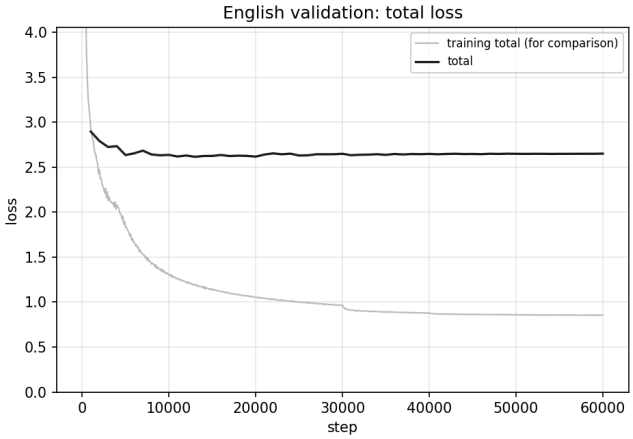
The reference trains LJSpeech for 900k steps. With 5 h of data, 60k steps is about 270 epochs for English and 400 for Telugu, and the validation curves below show that longer training would not help. The two models were trained one after the other: running both at once filled the GPU memory and cut throughput from 7–9 to 2.4 steps/s each.

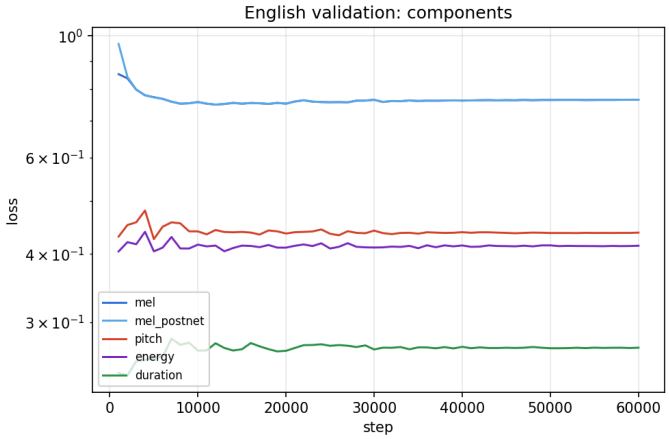
4. Training and validation loss



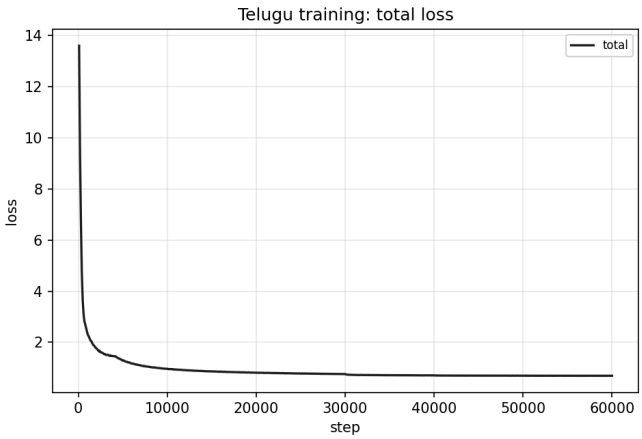


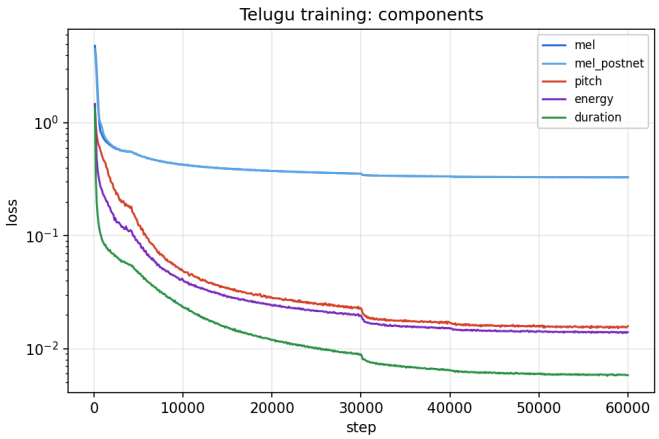
English training loss (mean over each 100-step window).



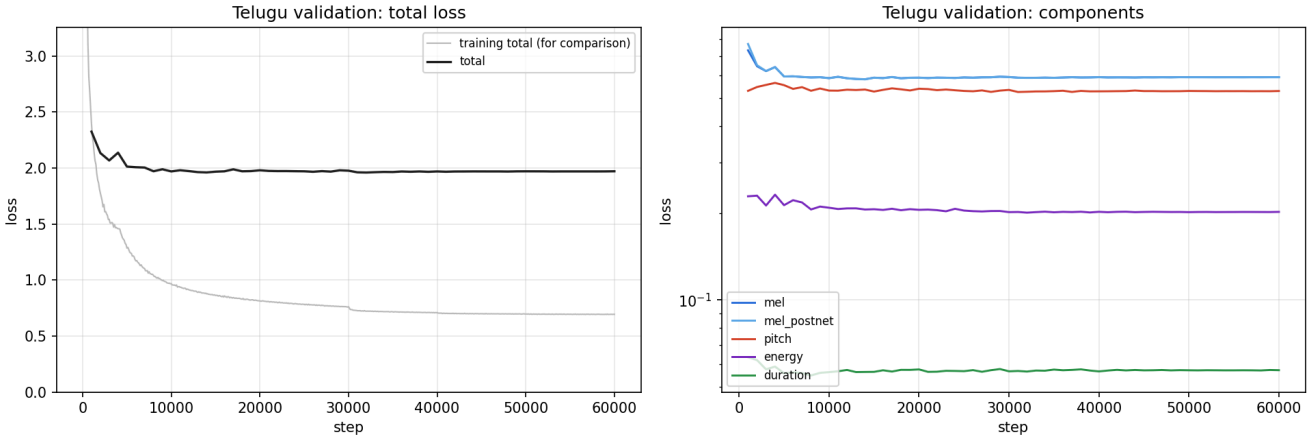


English validation loss (full validation set every 1,000 steps); the training total is drawn in grey for comparison.





Telugu training loss.



Telugu validation loss.

Loss at step	EN train 10k	EN train 60k	EN val best (13k)	EN val 60k	TE train 10k	TE train 60k	TE val best (32k)	TE val 60k
total	1.301	0.854	2.614	2.650	0.969	0.694	1.960	1.971
mel (post-net)	0.555	0.403	0.751	0.764	0.428	0.329	0.588	0.591
pitch	0.064	0.019	0.439	0.437	0.049	0.016	0.526	0.529
energy	0.070	0.018	0.404	0.414	0.041	0.014	0.201	0.202
duration	0.056	0.011	0.269	0.270	0.023	0.006	0.057	0.057

Convergence. Both runs are stable: no loss spikes, no divergence under bf16, and each learning-rate decay at 30k and 40k produces a visible step down in the training loss. Training loss falls fastest in the first 5k steps (warm-up ends at 4k) and is nearly flat after 50k.

Overfitting. Validation loss stops improving early: English reaches its minimum at 13k steps (2.614) and drifts up slightly to 2.650; Telugu is flat from about 10k steps (1.96–1.98) and reaches its minimum at 32k. The gap is largest in the variance predictors. The validation pitch loss is already at its final level after 1,000 steps (English 0.43, Telugu 0.53) while the training pitch loss keeps falling to 0.02, about 25 times lower. The predictors therefore memorise the training utterances' prosody rather than learn rules that transfer. The validation mel loss, which is computed with ground-truth durations, pitch and energy, improves until about 13k (English) and 20–30k (Telugu) and changes little afterwards. With 5 h of data, this is the expected behaviour of a 35 M-parameter model.

English vs Telugu. Every English validation loss is higher, most clearly duration (0.27 vs 0.057) and energy (0.41 vs 0.20). Svarah mixes 115 voices, recording conditions and spontaneous speech rhythm, and the pretrained US-English aligner fits Indian-accented speech less well than the Telugu aligner trained on the data itself, so English durations are both noisier and harder to predict.

Checkpoint choice. For Telugu the best-validation checkpoint (32k) was submitted; it sounds the same as the final one. For English the final checkpoint (60k) was submitted although its validation loss is 1.4 % higher than at 13k. In listening it is crisper (spectral contrast 18.1 vs 17.8 dB on the test sentences) and slightly faster, and the validation total is dominated by the pitch and energy terms, which hardly change between the two. Both checkpoints were rendered for comparison before this choice.

5. Generated speech

5.1 How the outputs were assessed

All ten test sentences were synthesised from the text exactly as given (including "rhyth m" and "అడ విలో"); the audio is in `Results/<language>/test_generated_audios/`. They were assessed by listening and by an automatic intelligibility check: `fastspeech2/asr_eval.py` transcribes the outputs with Whisper large-v3-turbo. It also scores 100 held-out

validation sentences, each synthesised in its own speaker's voice from the reference phone sequence, against the same ASR run on the real recordings of those sentences. The real-recording score shows how well the ASR model itself handles that speech. For Telugu the word error rate is uninformative (Whisper splits and joins Telugu words differently from the transcripts, giving 67 % WER even on real speech), so character error rate (CER) is the measure there.

100 validation sentences	synthesised WER	synthesised CER	real recordings WER	real recordings CER
English	18.2 %	9.8 %	5.7 %	2.8 %
Telugu	69.4 %	15.8 %	67.0 %	16.2 %

The Telugu model is as intelligible to the recogniser as the real recordings. In listening it sounds natural and expressive, with the lively, slightly theatrical delivery of professional dubbing artists (the IndicTTS voices). English synthesis loses clearly against real speech. With the speaker that has the most training data, it sounds muffled ("mumbly") but understandable.

5.2 Choice of the English voice

The English model has 115 learnt voices, and the voice has a large effect on intelligibility. All 115 were synthesised on 33 sentences (E1, E2a, E2b and 30 validation sentences) and transcribed. The WER on the validation sentences ranges from 8.7 % to 75 % (median 13.9 %), and it is unrelated to how much training audio the speaker has (correlation −0.04). The six best voices and the default were re-scored on 60 different validation sentences to avoid picking a voice that was lucky on the first set. `m_3834f1c1` (male, 3.9 min of training speech) was the most consistent (10.4 % and then 9.8 %), against 17.4 % and 20.6 % for `m_d270faea`, the speaker with the most training audio (15.3 min) that was used before. This voice is now the default for the English results.

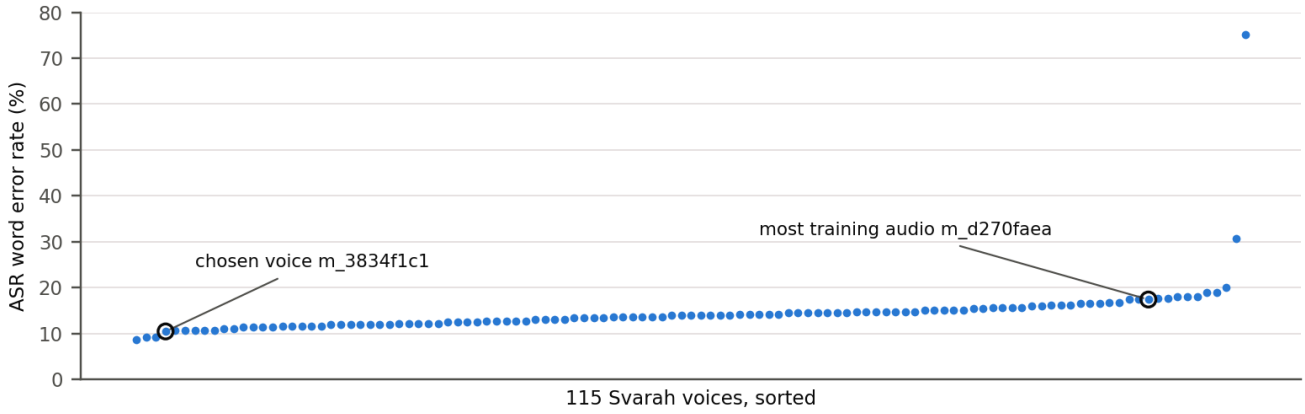


Figure 1. ASR word error rate of each English voice on 30 synthesised validation sentences.

5.3 Per-sentence observations

ID	What Whisper heard	Pronunciation and quality
E1	The anti-establishmentarianism movement gained considerable momentum during the parliamentary proceedings.	Everything but one syllable is correct. The 28-phone word is spoken in 2.1 s without a break; its "dis" syllable is reduced (the /ɪ/ lasts 35 ms) and is not heard. The front-end produced the correct full pronunciation.
E2a	The orchestra's conduct during the ceremony was praised by the audience.	Fully correct. "conduct" is given the noun pronunciation (K AA1 N D AH0 K T) by the part-of-speech tagger.
E2b	The researchers will conduct an experiment to evaluate the proposed method.	Fully correct; "conduct" gets the verb pronunciation (K AA0 N D AH1 K T). The acoustic stress difference is weak (section 6).
E3	The writing of the orchestra changed unexpectedly.	As required, "rhyth m" is read as written: "rhyth" is out of vocabulary and g2p_en predicts R AY1 DH ("rythe"), and the lone "m" is read as the letter name "em". "Rythe-em" is heard as "writing". The rest is correct.

E4	The meeting will start at 10.45 am on 12th November 2026. So and I should arrive on time.	Time, date and year are expanded and spoken correctly ("ten forty five A M", "twelfth November", "twenty twenty six"). "andaroo" (Telugu అందరూ, "everyone") is not English: g2p_en gives an anglicised /ænˈdɑroʊ/, heard as "and I". A short pause follows the comma.
T1	నియూరో† ఫార్మకో లాజికల్ పరిశోధనలో నూతనా ఉషధాల ప్రభావాన్ని విస్లిశిం చారు.	Intelligible and fluent; the long English loanword is pronounced in Telugu fashion. Errors: Whisper writes the loanword's first part "nyuu-ro" in Hindi (Devanagari) letters instead of Telugu; the palatal sibilant శ (sh) is heard as స (s) twice; ౌ (au) is heard as ఉ.
T2a	తాడు పయతని పట్టు చాలా బలంగా ఉంది.	Correct sounds; only the word boundary in "తాడుపై అతని" is placed differently, as happens in fluent speech.
T2b	ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.	Fully correct.
T3	పులి అడవిలో నిస్యబ్దంగా నడిచింది.	The split "అడ విలో" is read as written (no pause is inserted at a space), and the result is heard as the intended word అడవిలో. The geminate శ్శ is heard as "sy".
T4	ఈ సమావేసం రిండు వేల ఇరవయారు నవం బర్పన్నావ తేది ఉదయం పదిల భైయిదు గంటలకు ప్రారంభమవు తుంది. సమయానికి సిద్ధంగా ఉంచండి.	All numbers are spoken as Telugu words (రెండు వేల ఇరవై ఆరు, పన్నెండవ, పది నలభై ఐదు) and recognised as such; most differences are spelling and word-boundary variants. పన్నెండవ is reduced and శ again sounds like స. The comma produces a clear phrase break.

Underlined: words that differ from the input text. Whisper writes numbers as digits, so these count as correct for English. † Whisper wrote this word in Devanagari script; it is transliterated into Telugu here (the raw output is in `Results/Telugu/asr_eval.json`).

6. Variance-adaptor analysis

The figures below show, for each test sentence, the predicted duration of every phone (top; pauses grey), the predicted F0 (middle) and the predicted energy (bottom), with word boundaries as vertical lines. The separate full-resolution plots required by the assignment, which also overlay the contours on the generated mel spectrogram, are in `Results/<language>/variance_adaptor_plots/{duration,pitch,energy}/`. The raw values are in the JSON file next to them. F0 and energy are per-phoneme values, so each phone is one point.

6.1 Patterns common to all sentences

- **Declination and final fall.** Every sentence has its F0 peak early and ends low: English peaks at 130–157 Hz and ends at 86–97 Hz, Telugu peaks at 270–315 Hz and ends at 150–165 Hz. A declarative intonation contour is learnt in both languages.
- **Final lengthening.** The last phone of each sentence is among its longest (E2a /s/ 232 ms, E3 /i/ 255 ms, E4 /m/ 244 ms; Telugu final /i/ 139–163 ms), while energy drops to its lowest values. Telugu final vowels are almost devoiced (energy 10–26), as is common in Telugu statement endings.
- **Energy follows the phone class.** Vowels carry high energy (60–140), voiceless stops and fricatives are low (5–25) because stop closures are silent. The energy contour therefore alternates rapidly and mostly reflects segment identity rather than prosody.
- **Speaking rate** is 12–14 phones per second in both languages, about 80–100 ms per phone, close to the training data.
- **Pauses only at punctuation.** The front-ends insert a pause symbol only at punctuation, and the model predicts one pause per comma (E4 151 ms, T4 685 ms). Long sentences without commas (E1, T1) are spoken without any break.

6.2 English

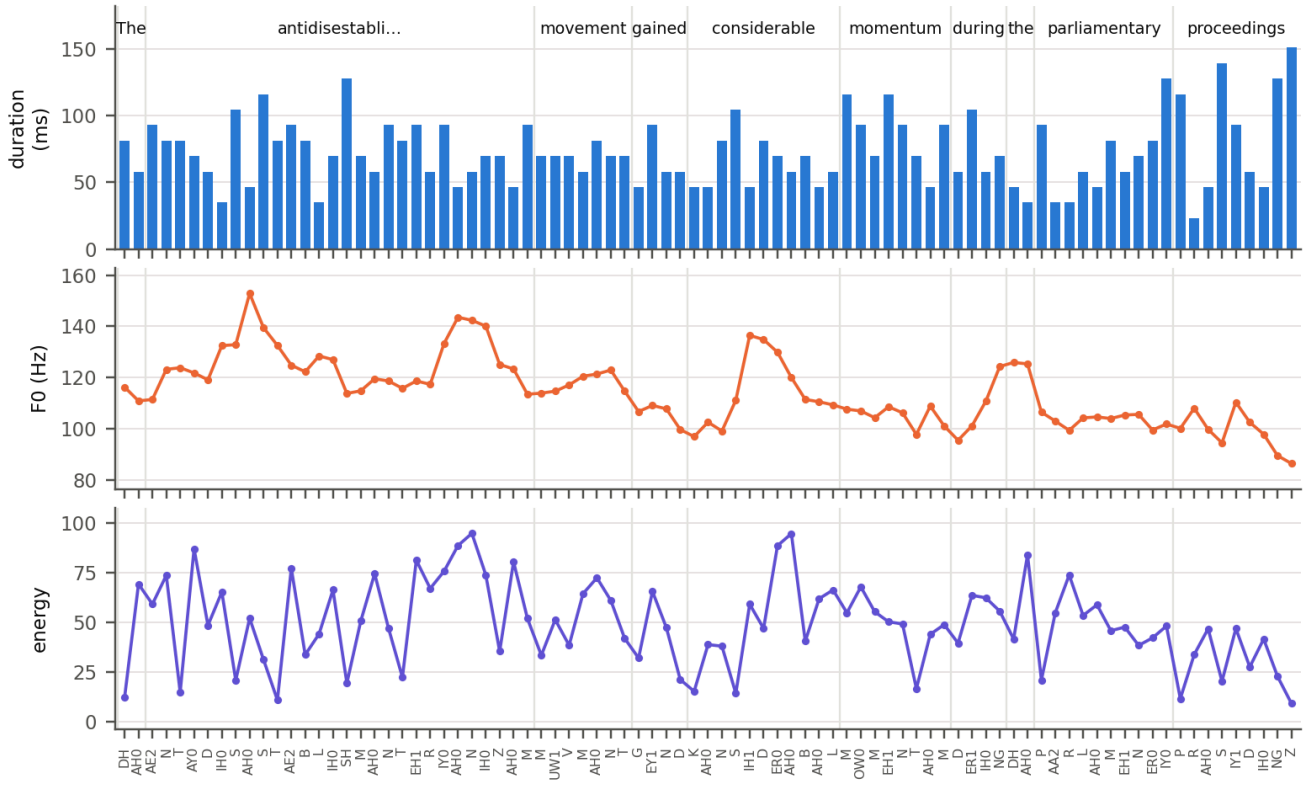


Figure 2. E1. The long word takes 2.1 s, 34 % of the sentence.

E1. "antidisestablishmentarianism" is handled as one prosodic unit with two local F0 peaks, 153 Hz on "-dis-es-" and 144 Hz on "-ar-i-an-". Most of its vowels are short (35–93 ms) and the sibilants long (/s/ 104–116 ms, /ʃ/ 128 ms); the very short /ɪ/ of "dis" (35 ms) explains why the syllable is lost. After the word, F0 declines steadily through "momentum … parliamentary" (about 105 Hz) to 86 Hz on the final /z/ of "proceedings", which with 151 ms is lengthened.

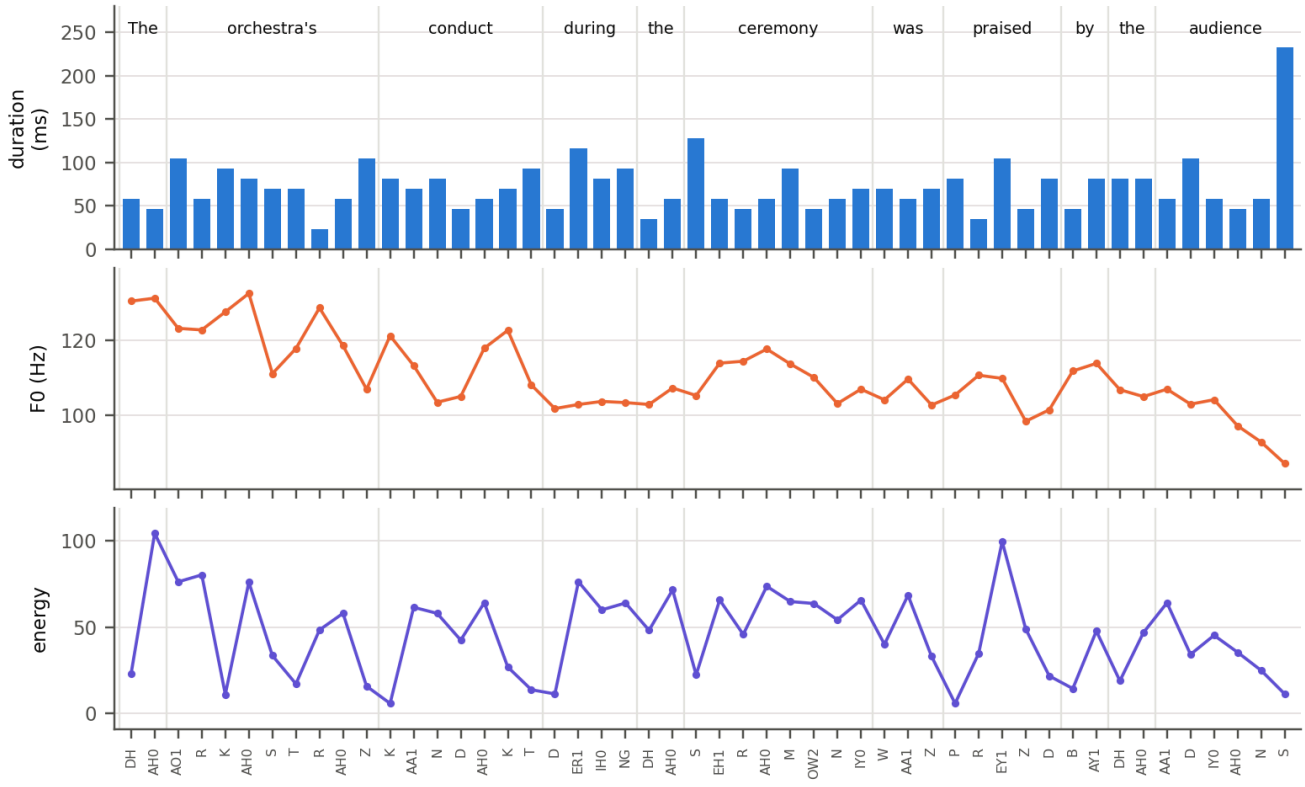


Figure 3. E2a, "conduct" as a noun.

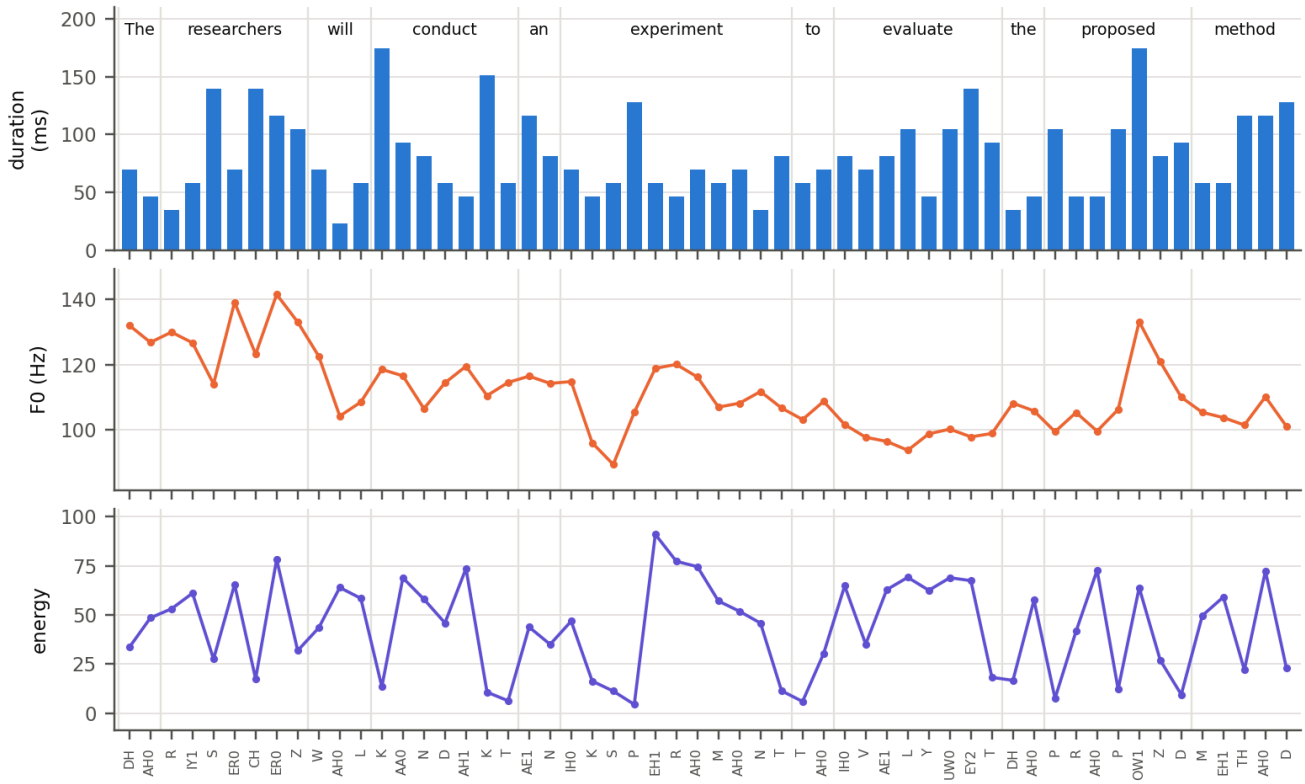


Figure 4. E2b, "conduct" as a verb.

E2a vs E2b (heteronym). The front-end gives the two "conduct"s different stress marks (AA1…AH0 vs AA0…AH1), and the adaptor treats them differently, but only weakly in the expected direction. In the noun, the first vowel is longer than the second (70 vs 58 ms) at similar F0 and energy. In the verb, the second (stressed) vowel has the higher energy (74 vs 69) and F0 (119 vs 117 Hz) but is shorter (46 vs 93 ms). The verb is longer as a whole (662 vs 499 ms), mostly because of long stop closures (/k/ 174 and 151 ms). English lexical stress is thus only partly realised. Two reasons are

plausible: Indian English, which Svarah represents, marks lexical stress less through vowel length and reduction than American English does, and stressed and unstressed variants of the same vowel are separate symbols that each see only a fraction of the 5 h of data. The sentence-level prosody is sound. E2a falls from 132 Hz on "orchestra's" to 87 Hz at the lengthened final /s/ of "audience" (232 ms). E2b has a high start on "researchers" (141 Hz) and a clear accent on "proposed" (vowel 174 ms, 133 Hz), i.e. the new information near the end of the sentence.

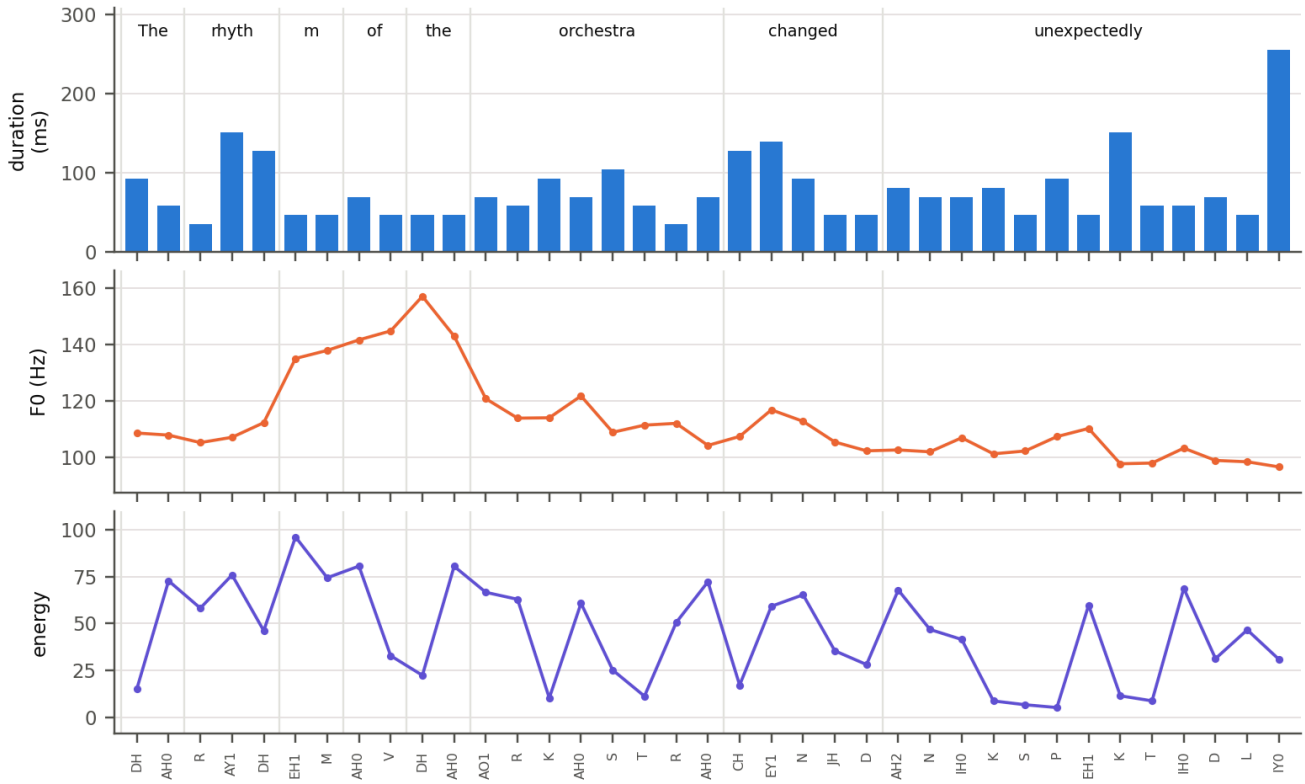


Figure 5. E3 with the split word "rhyth m".

E3. The split word becomes two words: "rhyth" = R AY1 DH (the diphthong lasts 151 ms) and the letter "m" = EH1 M. The adaptor treats the single letter like a spelled-out letter or acronym: F0 jumps from 108 to 136 Hz on "m" and keeps rising to 157 Hz over "of the", and "m" gets the highest energy of the sentence (96). This prominence on a function-word region is unnatural and is a direct effect of the input text. From "orchestra" on, the contour is normal: declination to 96 Hz and strong final lengthening of the last /i/ of "unexpectedly" (255 ms).

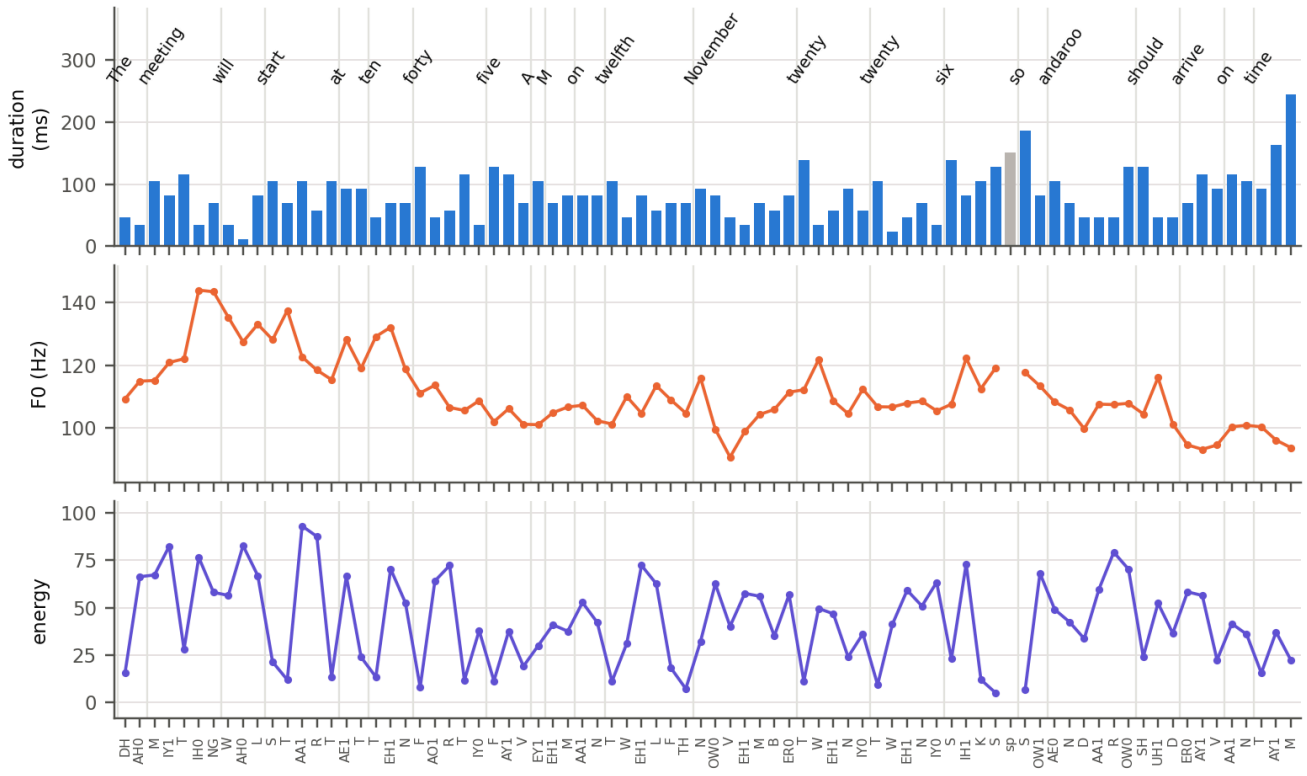


Figure 6. E4 with the normalised time, date and year.

E4. The highest F0 (144 Hz) is on "meeting", followed by declination through the time expression, which is spoken as "ten forty five A M" with the two letters as separate words (104 and 151 ms). The pause after the comma lasts 151 ms; F0 resets only slightly after it (to 118 Hz on "so"), and the second phrase falls to 93 Hz on "arrive" with strong final lengthening on "time" (/m/ 244 ms). "andaroo" (AE0 N D AA1 R OW0) gets F0 around 107 Hz with its highest energy on the /r/, an ordinary unaccented English word.

6.3 Telugu

Two properties of Telugu are clearly learnt. **Vowel length** is phonemic, and long vowels are predicted 1.5–3 times longer than short ones (e.g. *aa* 139–163 ms vs *a* 35–58 ms in T2a). **Geminate consonants** are realised as one long closure: *ṭṭ* in పట్టు lasts 116–139 ms, *cc* in వచ్చింది 163 ms, *śś* in నిశ్శబ్దంగా 209 ms, against 45–90 ms for single stops. A third pattern appears in every sentence: non-final words end with a **rise** on their final vowel (continuation), and only the final verb **falls**. Examples are పట్టు ending at 257/290 Hz, రెండు at 264 Hz and గంటలకు at 249 Hz, while ఉంది, వచ్చింది, నడిచింది, ఉంచండి fall to 150–165 Hz. This is the typical intonation of Telugu declaratives.

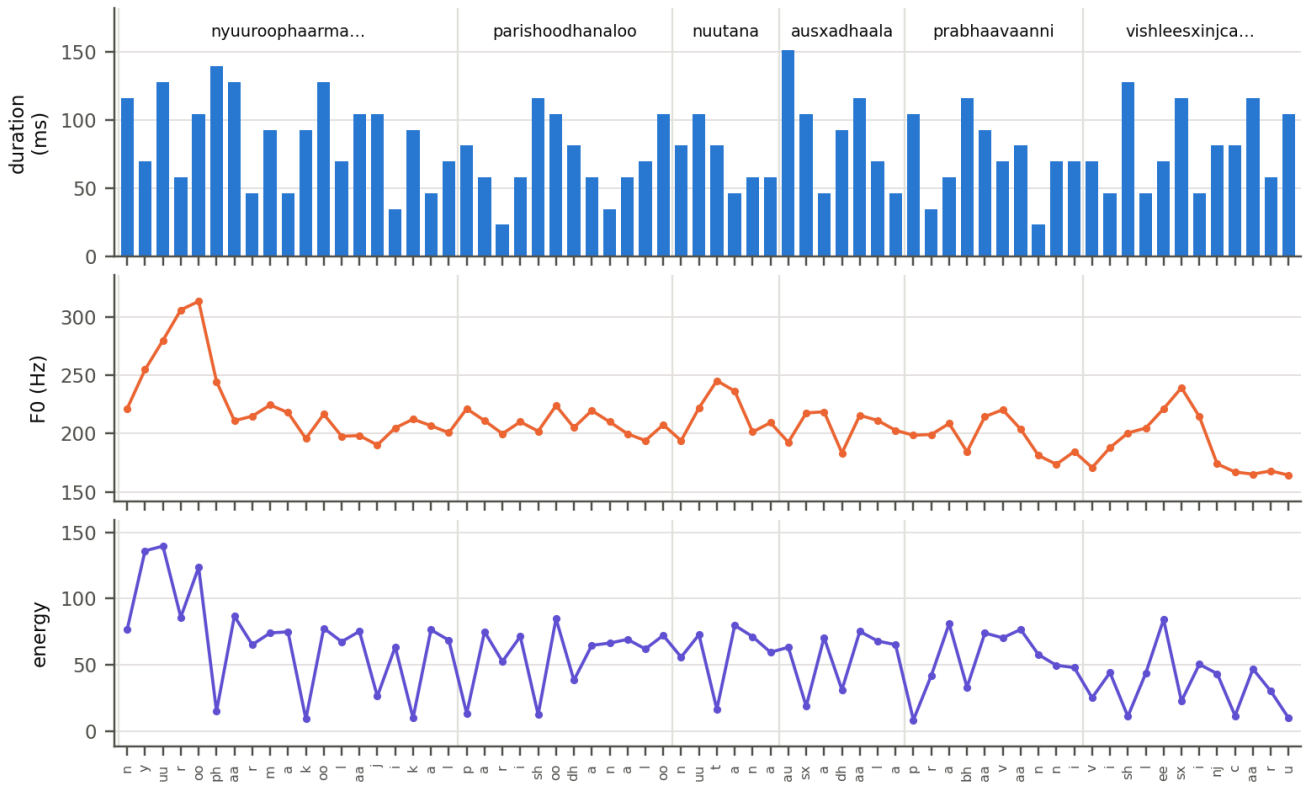


Figure 7. T1 (romanised phone labels; x marks retroflex consonants).

T1. The loanword న్యూరోఫార్మకోలాజికల్ takes 1.67 s and carries the sentence's F0 peak (313 Hz on "roo") right at its start, after which F0 drifts down over the following words (word means 227, 209, 218, 206, 197, 190 Hz). The sentence-final విశ్లేషించారు falls to 164 Hz with an almost voiceless final /u/ (energy 10). The anusvara before చ becomes the palatal nasal (*ɲ*), as the G2P rule intends.

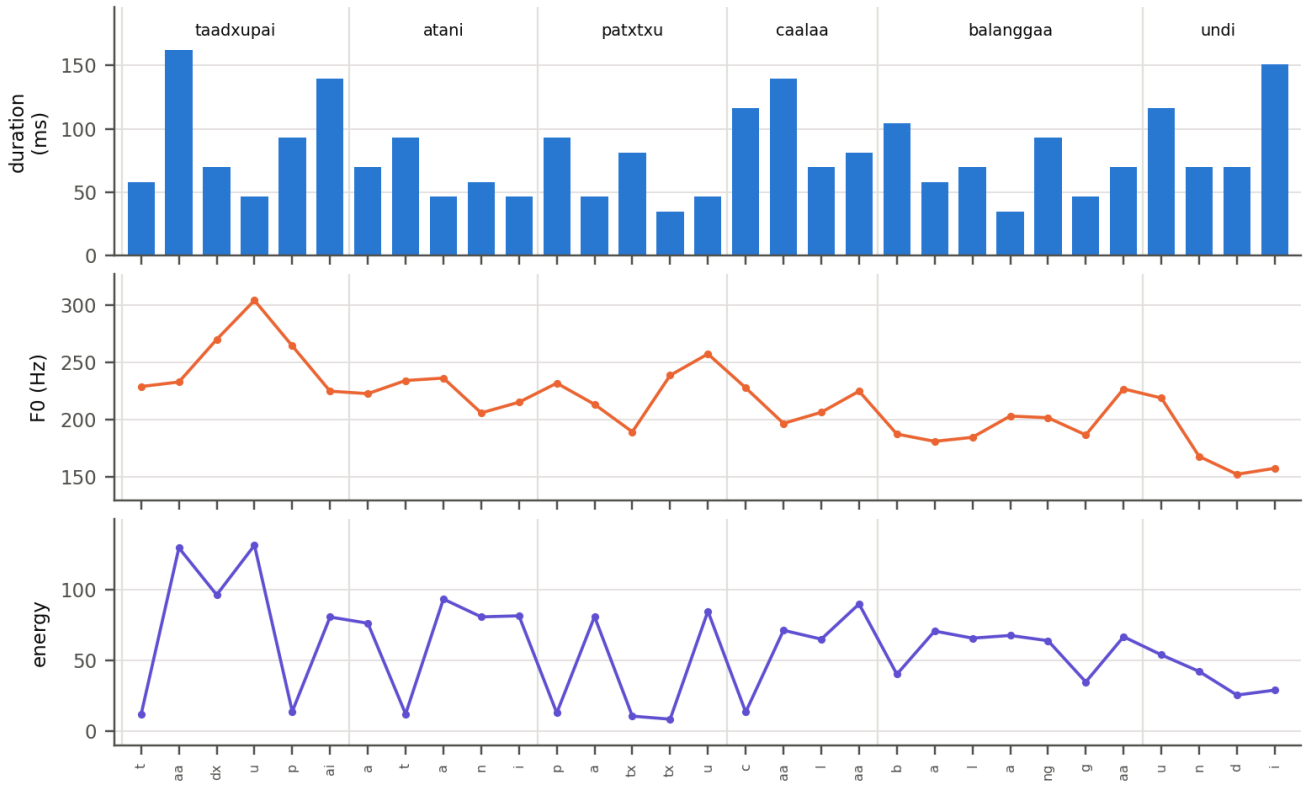


Figure 8. T2a, పట్టు = grip.

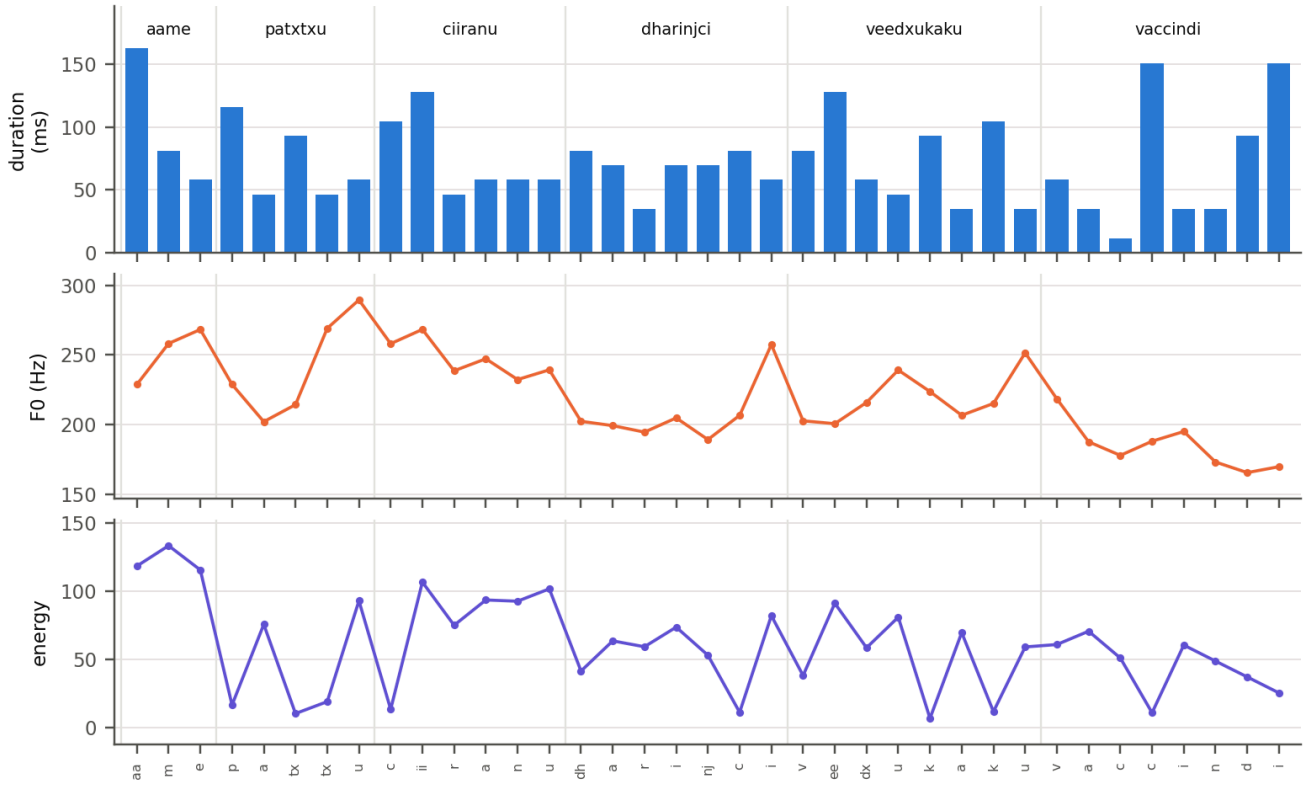


Figure 9. T2b, పట్టు = silk.

T2a vs T2b (homograph). పట్టు means "grip" in T2a and "silk" in T2b, but unlike English "conduct" the two senses are pronounced identically in Telugu, so the same phones are correct for both. The predictions are indeed very similar: 302 vs 360 ms, both with a long geminate closure and a rise to the final /u/. The rise is higher in T2b (290 vs 257 Hz), most likely because there the word comes right after the high-pitched sentence opening ఆమె (268 Hz). FastSpeech 2 sees no meaning, only phones and their context, so position is the only source of difference. T2a has the clearest example of the continuation rise: తాడుపై peaks at 304 Hz, before declination and the final fall on ఉంది (219 → 152 Hz).

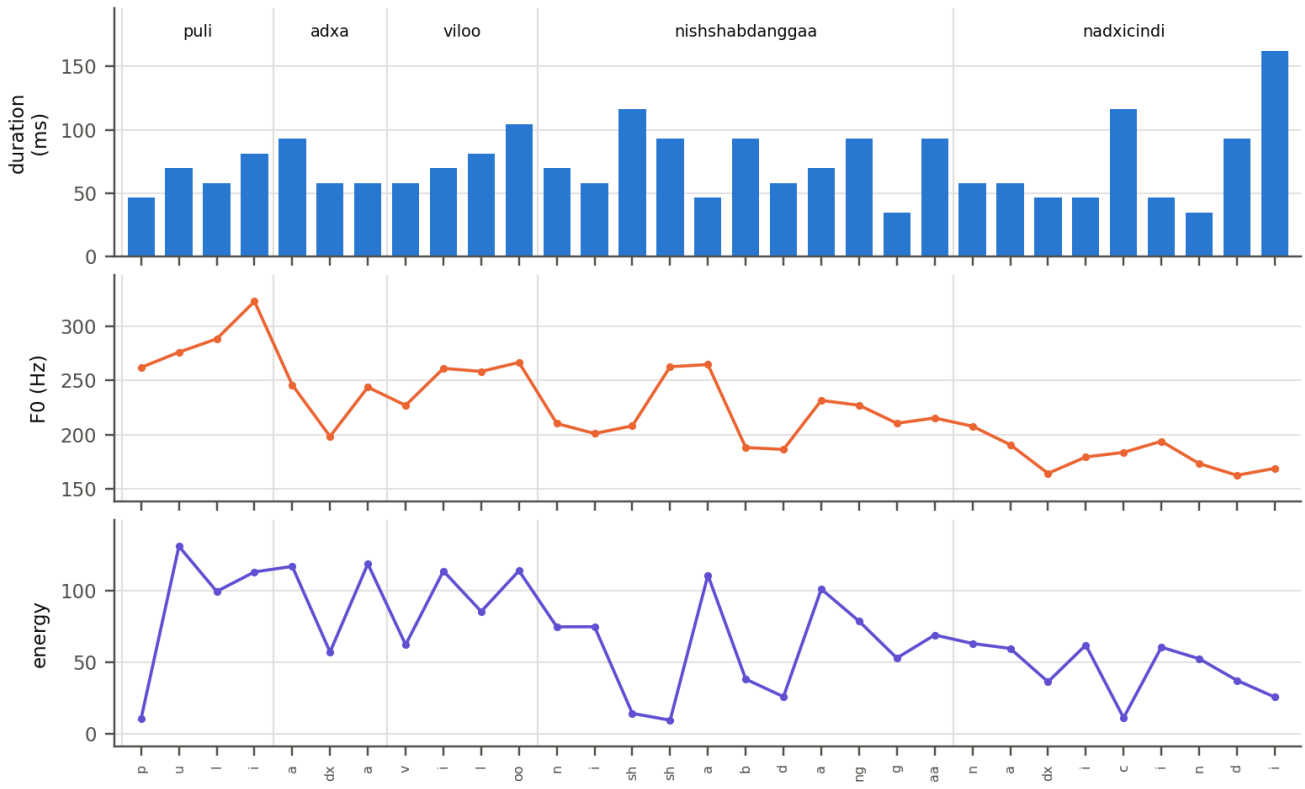


Figure 10. T3 with the split word "అడ విలో".

T3. Because a space is not punctuation, "అడ విలో" is synthesised as two consecutive words without a pause (209 + 313 ms). The boundary is hardly marked: F0 dips at the retroflex /ɖ/ (198 Hz) and continues into "vi" (261 Hz), so the listener (and the ASR) hears అడవిలో. Telugu's consonant-vowel syllable structure makes the split harmless here, unlike "rhyth m" in English. The sentence opens very high on పులి (323 Hz on the final /i/) and ends with the usual fall on నడిచింది (163–169 Hz, final /i/ 163 ms).

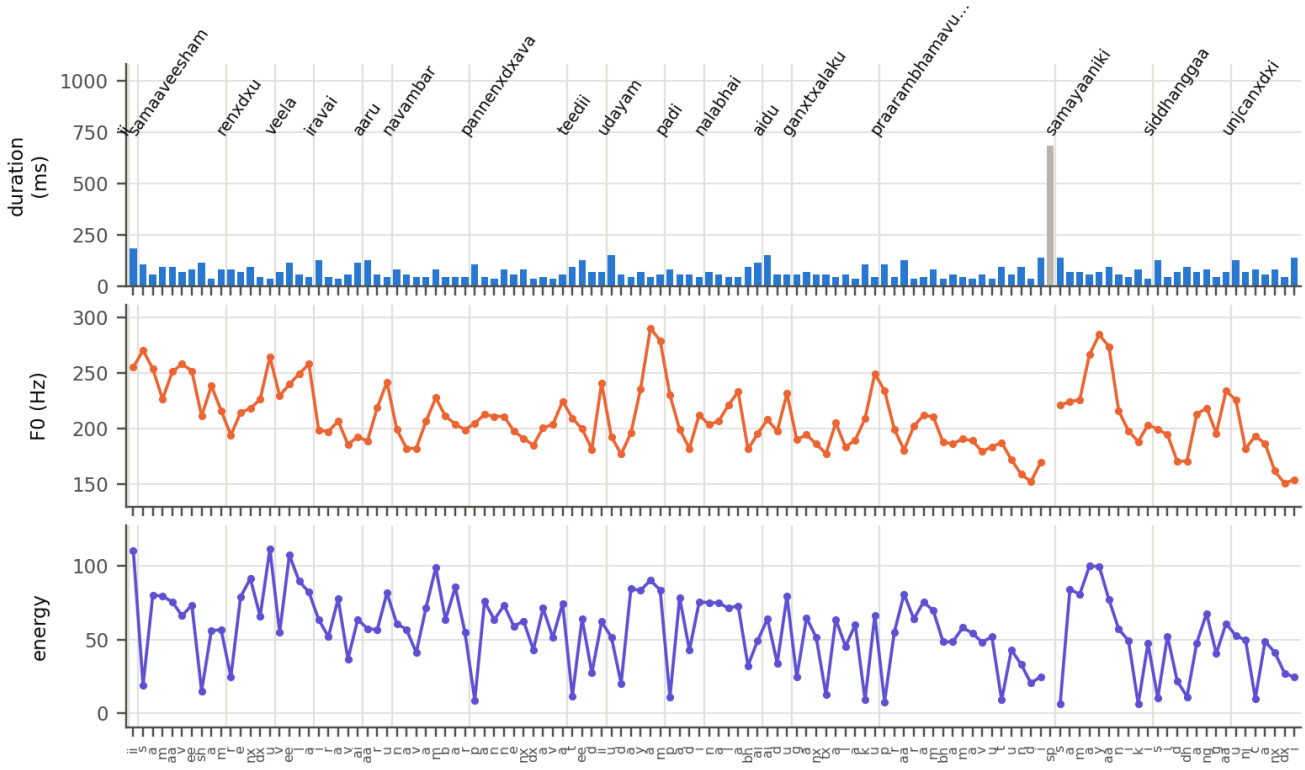


Figure 11. T4 with the normalised year, ordinal date and time.

T4. The longest sentence (9.2 s, 119 phones) is also the fastest (14 phones/s). The number words (రెండు వేల ఇరవై ఆరు, పన్నెండవ, పది నలభై ఐదు) are integrated into the phrase like ordinary words, each ending with a small rise. The comma produces a 685 ms pause and splits the sentence into two intonation phrases. The first ends with a fall to 152 Hz on ప్రారంభమవుతుంది; after the pause F0 is reset to 284 Hz on సమయానికి and falls again to 150 Hz on ఉంచండి. This pitch reset is the clearest phrase-level structure in all ten sentences. ఉదయం carries a local peak (290 Hz), a focus-like accent on the time of day.

7. Comparison, limitations and possible improvements

7.1 Why Telugu works better

- **Data per voice**: Telugu has 2.5 h of studio speech per professional speaker; English has a median of 1.9 minutes per speaker, recorded on varied devices and partly spontaneous. A multi-speaker model averages over many voices and recording conditions, which gives the muffled English sound.
- **Orthography**: Telugu script maps almost one-to-one to sounds, so a short rule set gives nearly error-free phones. English needs a lexicon, a statistical G2P for out-of-vocabulary words, and stress, which the model realises only partly.
- **Alignment**: the Telugu aligner was trained on the same two voices; the English one is a US-English model applied to Indian-accented speech. Noisier English durations show up as a five times higher validation duration loss.

7.2 Failures observed on the test sentences

- Split words (E3, T3) are synthesised as written, by requirement. This is harmless in Telugu but produces "rythe em" in English, with a spurious accent on the letter "m".
- Code-mixed words: "andaroo" (E4) is anglicised because the English model has no Telugu phones and g2p_en guesses an English spelling-to-sound mapping.
- Weak lexical stress contrast in English (E2a/E2b), and a reduced syllable in a very long word (E1).
- Telugu శ /ʃ/ is repeatedly recognised as /s/ (T1, T3, T4). It is unclear whether the model under-articulates it or the ASR confuses the two; it is the most consistent Telugu error.
- No pauses in long sentences without punctuation (E1, T1), because pauses come only from punctuation.
- Both models' variance predictors overfit after a few thousand steps (section 4).

7.3 Possible improvements

- **English data**: train on fewer speakers with more data each (e.g. the 10–20 most intelligible voices), filter noisy clips with an SNR or ASR-confidence threshold, or pretrain on a large clean corpus (LJSpeech, LibriTTS) and fine-tune on Svarah to keep the Indian accent.
- **Prosody**: normalise F0 per speaker instead of over the whole corpus (much of English pitch variance is between male and female voices); early stopping or stronger regularisation for the variance predictors; a pause/phrase-break predictor for long sentences.
- **Front-ends**: a Telugu-aware path for Telugu words in English text (transliteration into the English phone set); a small Indian-English lexicon; Telugu loanword rules.
- **Telugu data**: most of the 36.7 h corpus (mainly long female paragraphs) was excluded by the 15 s limit; cutting those recordings at aligned pauses would add data at no recording cost.
- **Vocoder**: fine-tune HiFi-GAN on each language's ground-truth-aligned mel spectrograms to remove the slight metallic quality of the universal vocoder.

8. Reproducibility

Software (pinned in `requirements.txt`): Python 3.10, PyTorch 2.11 (CUDA 12.8 wheels, required for RTX 50-series GPUs), NumPy 2.2, librosa 0.11, pyworld 0.3.5, tgt 1.5, g2p-en 2.1, NLTK 3.10, scikit-learn 1.7, matplotlib 3.10, PyYAML, TensorBoard; Montreal Forced Aligner 3.4.3 in a separate conda environment; transformers 5.20 and jiwer 4.0 only for the ASR evaluation. Seeds: 1234 for data selection and training, 0 for the ASR sample.

Configuration: `English/config.yaml` and `Telugu/config.yaml` (paths, selection quotas, audio/feature settings, model and training hyper-parameters). Corpora are expected next to the repository in `../data/`, intermediate files go to `../work/`.

```
conda create -n fs2 python=3.10 && conda activate fs2
pip install torch==2.11.0 torchaudio==2.11.0 --index-url https://download.pytorch.org/whl/cu128
pip install -r requirements.txt
conda create -n mfa -c conda-forge montreal-forced-aligner=3.4.3 python=3.11
conda run -n mfa mfa model download acoustic english_us_arpa

huggingface-cli download SPRINGLab/IndicTTS_Telugu --repo-type dataset --local-dir ../data/IndicTTS_Telugu
huggingface-cli download ai4bharat/Svarah --repo-type dataset --local-dir ../data/Svarah   # gated: log in
first

cd Telugu            # or English; identical commands
python preprocess.py prepare     # selection, 22.05 kHz wav + transcripts, lexicon
python preprocess.py align       # MFA
python preprocess.py features    # mel / pitch / energy / duration, train.txt, val.txt
python train.py                  # 60k steps, then Results/<language>/*_loss.png
python train.py --export best    # Telugu (English: --export 60000) -> checkpoints/final.pth.tar
python inference.py              # test sentences -> Results/<language>/
python ../fastspeech2/asr_eval.py Telugu    # optional intelligibility check
```

Checkpoints: `English/checkpoints/final.pth.tar` (step 60,000) and `Telugu/checkpoints/final.pth.tar` (step 32,000) hold the model weights in float16 (72 MB each, under GitHub's 100 MB file limit; the training checkpoints with optimiser state are 418 MB). Inference with them reproduces the submitted audio; compared with the float32 weights the predicted F0 differs by less than 0.1 Hz.

References

1. Y. Ren, C. Hu, X. Tan, T. Qin, S. Zhao, Z. Zhao, T.-Y. Liu. FastSpeech 2: Fast and High-Quality End-to-End Text to Speech. ICLR 2021 (arXiv:2006.04558).
2. C.-M. Chien (ming024). FastSpeech2: a PyTorch implementation of FastSpeech 2. github.com/ming024/FastSpeech2.
3. J. Kong, J. Kim, J. Bae. HiFi-GAN: Generative Adversarial Networks for Efficient and High Fidelity Speech Synthesis. NeurIPS 2020.
4. M. McAuliffe et al. Montreal Forced Aligner. Interspeech 2017; version 3.4.3.
5. T. Javed et al. Svarah: Evaluating English ASR Systems on Indian Accents. Interspeech 2023.
6. Indic TTS consortium, IIT Madras. IndicTTS database (Telugu); Hugging Face copy SPRINGLab/IndicTTS_Telugu.
7. A. Radford et al. Robust Speech Recognition via Large-Scale Weak Supervision (Whisper). ICML 2023.